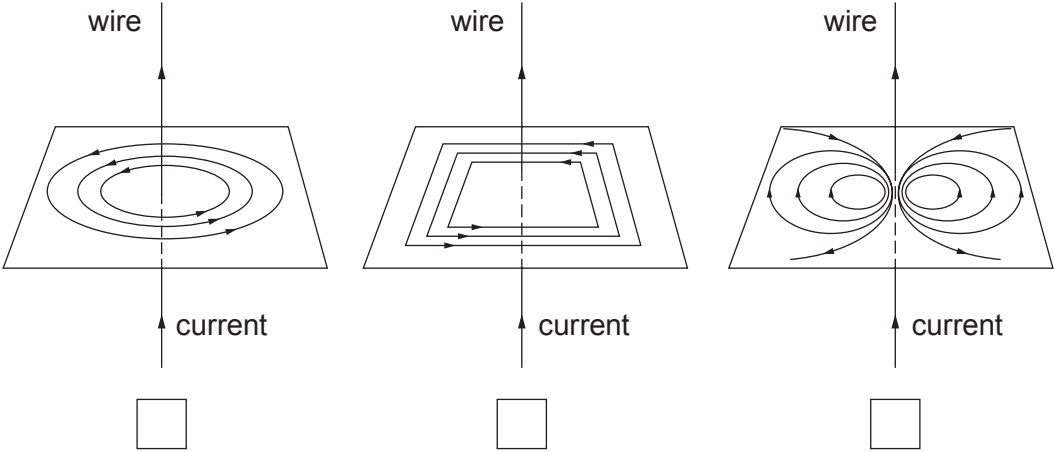
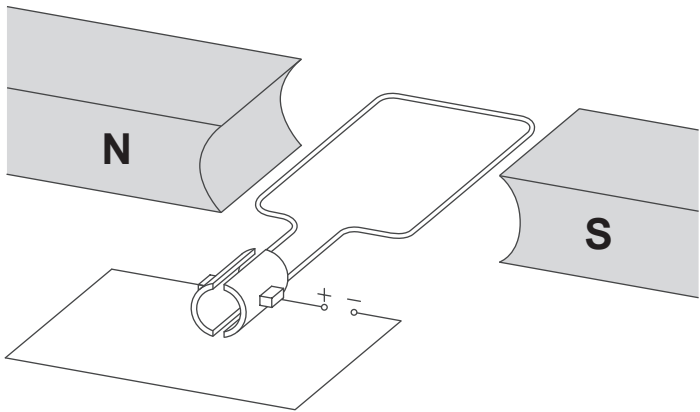


4. (a) Tick (✓) the box below the diagram which correctly shows the magnetic field pattern around a current-carrying straight wire. [1]



(b) The diagram below shows a simple electric motor. When there is a current in the coil it experiences a force due to the magnetic field and starts to spin.



(i) State **one** way in which the coil could be made to spin in the opposite direction. [1]

.....

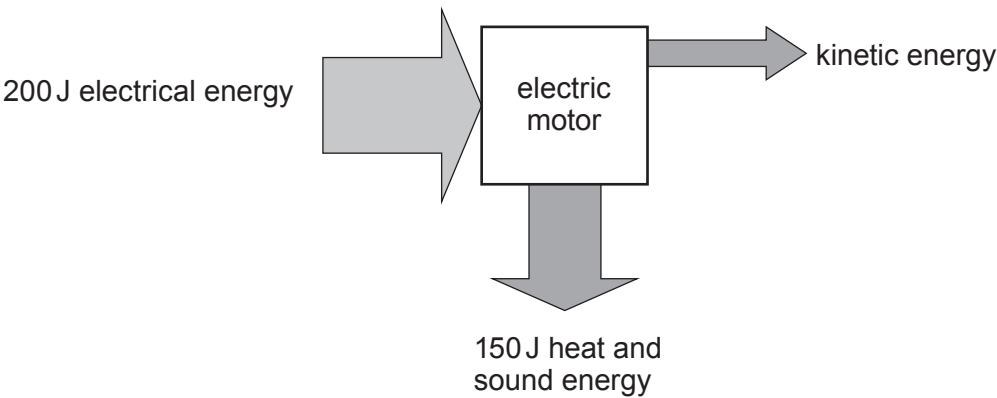
(ii) State **two** ways in which the coil could be made to spin faster. [2]

1.

2.



(c) A simple electric motor transfers energy as shown in the Sankey diagram below.



(i) Calculate how much useful energy the motor produces. [1]

Useful energy = J

(ii) Use an equation from page 2 to calculate the % efficiency of the motor. [2]

% efficiency =

7

